Extension: Chords of a circle

1 $x = 32$

2

a $x = 60^\circ$ **b** $x = 54^\circ$ **c** $x = 23$ **d** $x = 21$

e $x = 10$ **f** $x = 16$

3 $AB = 16 \text{ cm}$

4 $x = 33^\circ$

5

a $OA = \sqrt{205}$ **b** $BP = \sqrt{530}$

6 $XN = 9 \text{ cm}$

7 $YZ = 19 \text{ cm}$

8 $SV = 26.25 \text{ cm}$

9 $r = 17 \text{ cm}$

10

To prove: $\triangle ABO \cong \triangle CDO$

	Statements	Reasons
1.	$AB = CD$	Given
2.	$m\angle OEB = m\angle OFD = 90^\circ$	Given
3.	$\overline{AO} \cong \overline{OB} \cong \overline{OD} \cong \overline{OC}$	Definition of radii
4.	$\overline{AB} \cong \overline{CD}$	Definition of congruent segments
5.	$\triangle ABO \cong \triangle CDO$	SSS congruence theorem

11

To prove: $\triangle$[redacted] $\cong \triangle OBC$

	Statements	Reasons
1.	$\overline{OC} \perp \overline{AB}$	Given
2.	$\overline{OA} \cong \overline{OB}$	Definition of radii
3.	$m\angle ACO = \angle BCO = 90^\circ$	Definition of perpendicular segments
4.	$\overline{OC} \cong \overline{OC}$	Reflexive property
5.	$\triangle OAC \cong \triangle OBC$	RHS congruence theorem

12

To prove: $\overline{AB} \cong \overline{DC}$

	Statements	Reasons
1.	$\angle AOB \cong \angle DOC$	Given
2.	$m\angle AOB = m\angle DOC$	Definition of congruent angles
3.	$\overline{OA} \cong \overline{OB} \cong \overline{OC} \cong \overline{OD}$	Definition of radii
4.	$m\angle ACO = m\angle BCO = 90^\circ$	Definition of perpendicular segments
5.	$\triangle AOB \cong \triangle DOC$	SAS congruence theorem
6.	$\overline{AB} \cong \overline{DC}$	CPCTC

13

To prove: $\triangle ABO \cong \triangle DEO$

	Statements	Reasons
1.	$\overline{AB} \parallel \overline{ED}$	Given
2.	$\overline{OA} \cong \overline{OD}$	Definition of radii
3.	$\angle BAO \cong \angle EDO$	If two lines are parallel, then the alternate interior angles are congruent (Alternate interior angles theorem)
4.	$\angle AOB \cong \angle DOE$	Vertical angles theorem
5.	$\triangle ABO \cong \triangle DEO$	ASA congruence theorem

Additional questions

14

a $x = 4$ **b** $m\overset{\frown}{RS} = 40^\circ$ **c** $m\overset{\frown}{RQ} = 88^\circ$ **d** $x = 3$

15 $m\overset{\frown}{BC} = 118^\circ$

16

a True **b** False **c** False **d** False

17 30 feet

18

a $y = 7$ **b** $UV = 11 \text{ cm}$

19

a $x = 16$ **b** $m\angle SPT = 160^\circ$

20

a $x = 7$ **b** $MN = 62$

21

a Since chords $\overline{AD}$ and $\overline{EB}$ are congruent, their corresponding central angles are congruent.

b $m\overset{\frown}{AD} = 122^\circ$

c $m\overset{\frown}{DB} = 76^\circ$

22 The segment length of eyebrow which cuts through the left eye is equal in length to that which cuts through the right eye.

23

a $x = 6$ **b** $m\angle 1 = 49.5^\circ$ **c** $m\overset{\frown}{EDC} = 125^\circ$ **d** $m\angle 2 = 61.5^\circ$

24

To prove: $\overset{\frown}{AB} \cong \overset{\frown}{DC}$

	Statements	Reasons
1.	$\angle AOB \cong \angle DOC$	Given
2.	$\overline{AO}$, $\overline{OC}$, $\overline{OD}$, and $\overline{OB}$ are radii of circle O	Definition of radii
3.	$\overline{AO} \cong \overline{OB} \cong \overline{OD} \cong \overline{OC}$	Definition of radii
4.	$\triangle AOB \cong \triangle DOC$	SAS congruence theorem
5.	$\overline{AE} \cong \overline{BD}$	CPCTC
6.	$\overset{\frown}{AB} \cong \overset{\frown}{DC}$	If two chords in a circle are congruent, then their corresponding arcs are congruent

25

To prove: $\overline{JP}$ bisects $\overset{\frown}{KJM}$

	Statements	Reasons
1.	circle P	Given
2.	$\overline{KM} \perp \overline{JP}$	Given
3.	$\overline{KP} \cong \overline{PM}$	Definition of radii
4.	$\angle PLK \cong \angle PLM$	Definition of right angles
5.	$\overline{PL} \cong \overline{PL}$	Reflexive property
6.	$\triangle PLK \cong \triangle PLM$	SAS congruence theorem
7.	$\angle MPJ \cong \angle KPJ$	CPCTC
8.	$\overset{\frown}{KJ} \cong \overset{\frown}{JM}$	If two central angles are congruent, then their corresponding arcs are congruent
9.	$\overline{JP}$ bisects $\overset{\frown}{KJM}$	Definition of bisector

26

Since $\overset{\frown}{BC} \cong \overset{\frown}{ED}$, we know that $\overline{BC} \cong \overline{ED}$. Since $\angle CBE$ and $\angle EDC$ are inscribed in the same arc, they are congruent. Since $\angle BCE$ and $\angle DCE$ are inscribed in congruent arcs, they are also congruent. This means by AAS, $\triangle BCE$ and $\triangle DEC$ are congruent, so $\overline{CD} \cong \overline{EB}$ by CPCTC